RCS Day 1 - Residential Systems & Shop Safety

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Architecture of the Day

- Theme: Residential construction is a set of connected systems: structure, electrical, plumbing, envelope, and finish.
- Core idea: Students begin RCS by learning how professionals move safely, account for tools, and verify work before trusting it.
- Primary lab: Rolling sink module - assemble, inspect, leak test, troubleshoot, document, and reset.
- Capstone thread: Today introduces the systems thinking and safety behavior needed for each student's individual wall project.
- Room tone: Calm, practical, low-drama, one task at a time. No shame, no rushing, no guessing.

Learning Targets

- Identify the major systems in a residential building.
- Select appropriate PPE for common shop and lab tasks.
- Explain stop-work authority and tool accountability.
- Identify common power tools and their primary hazards.
- Assemble a simple sink drain and P-trap system in a rolling lab module.
- Perform a leak inspection, correct issues, and document verified work.

Materials

- Safety glasses
- Hearing protection
- Gloves for material handling, not rotating tools
- Tape measures
- Pencils/markers
- PPE checklist
- Tool checkout sheet
- Drill/driver, impact driver, circular saw, miter saw, jigsaw, oscillating tool for demonstration
- Rolling sink lab modules
- Faucets, sink basins, tailpieces, P-traps, slip nuts, washers, drain pipe sections
- Buckets/towels
- Student packets
- Verification logs

- Exit tickets

Daily Schedule

Time	Block	Instructor Action
10:00-10:10	Arrival + setup	Students get notebooks, pencils, quiz sheet. Instructor sets out PPE and rolling sink modules but keeps water off.
10:10-10:20	Safety reset + day goal	Frame RCS as residential systems plus professional habits. Tool use is earned by demonstrated control.
10:20-10:35	Morning quiz	Individual, quiet. Use as a teaching check, not a trap.
10:35-10:50	Quiz review	Correct together. Ask students to explain why a safety choice matters.
10:50-11:15	Slides part 1	RCS overview, building systems, final wall capstone preview.
11:15-11:45	Safety + PPE demo	PPE selection, stop-work authority, tool accountability.
11:45-12:05	Power tool intro	Drill/driver, circular saw, miter saw, jigsaw, oscillating tool. Demonstration only unless students are cleared.
12:05-12:30	Whiteboard challenge	Map a simple sink system: supply -> shutoff -> faucet -> sink -> drain -> P-trap -> outlet.
12:30-1:00	Lunch / reset	No unsupervised tool or water work during lunch.
1:00-1:15	Lab demo	Show parts, washer orientation, hand-tight assembly, leak-test setup, and documentation expectation.
1:15-2:15	Rolling sink lab	Teams inventory, assemble, inspect, leak test, correct, and document.
2:15-2:35	Troubleshooting + retest	Groups diagnose leaks or alignment problems, then retest.
2:35-2:50	Cleanup + evidence	Water off, station dry, parts counted, tools returned, verification log completed.
2:50-3:00	Exit ticket + dismissal	Students answer four prompts and preview Day 2 measurement/framing.

Opening Script

- Welcome students by name when possible. Keep the first five minutes low pressure.
- Say: "Today starts Residential Construction Specialty. We are moving from solar systems into building systems, but the rule stays the same: we verify before we trust."
- Say: "Tools do not care how confident you feel. They care where your hands, eyes, material, and attention are."
- Say: "The quiz tells me what I need to reteach. It is not a trap."

Power Tool Introduction Boundary

- Today introduces common shop power tools. It does not automatically clear students for independent use.
- A student earns tool use through demonstration: PPE, inspection, setup, body position, controlled operation, shutdown, and cleanup.

- No student operates a saw independently on Day 1 unless the instructor has directly cleared that student and the task.

Required Stop Conditions

- Any student is unsure of the next step.
- A tool is damaged, missing a guard, or behaving oddly.
- PPE is missing or inappropriate for the task.
- A cut path, bit path, cord path, or water path is unclear.
- Hands, clothing, jewelry, or another person are in the danger zone.
- Water leaks onto a walking surface or near electrical equipment.
- The group cannot explain the system path before testing.

Afternoon Lab: Rolling Sink Module

1. Assign roles: Builder, Inspector, Documenter. Rotate if time allows.
2. Inventory parts and identify each component's purpose.
3. Dry-fit the assembly before tightening.
4. Assemble by hand first; avoid overtightening plastic fittings.
5. Stop for inspection before water testing.
6. Leak test slowly with bucket/towels ready.
7. Diagnose any leak: alignment, washer, slip nut, missing part, or overtightening.
8. Correct one issue at a time, then retest.
9. Document verified work and unresolved questions.
10. Disassemble/reset station only after instructor approval.

Common Mistakes to Watch For

- Cross-threaded slip nut
- Missing or reversed washer
- Trap arm out of alignment
- Overtightening plastic fittings
- No towel/bucket before testing
- Repair made without documenting the leak first
- Students crowding the module instead of using roles

End-of-Day Closeout

- All water off.
- Modules dry.
- Tools and parts counted.
- Floor clear of slip/trip hazards.
- Student verification logs collected or checked.
- Exit tickets collected.

- Instructor notes captured for Day 2 adjustments.